



DS & AI

Machine Learning

Unsupervised Learning
DPP 02 Discussion Notes



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#Q. Consider the following Confusion Matrix:

	Predicted A	Predicted B	Predicted C
Actual A	40	5	5
Actual B	10	30 ✓	10
Actual C	5	10	35



Precision = 0.73

Recall = 0.8 $\rightarrow \left(\frac{TP}{TP+FN} \right) \Rightarrow TP/Total\ Actual\ P$

$\nearrow \frac{TP}{FP+TP} \Rightarrow TP/Total\ Pred.\ P$



Precision = 0.73

Recall = 0.76



Precision = 0.76

Recall = 0.81



Precision = 0.8

Recall = 0.8

$$P_A = \frac{40}{55}$$

$$P_B = \frac{30}{45}$$

$$P_C = \frac{35}{50}$$

$$\left. \begin{matrix} P_A = \frac{40}{55} \\ P_B = \frac{30}{45} \\ P_C = \frac{35}{50} \end{matrix} \right\} \quad R_A = \frac{40}{50}, R_B = \frac{30}{50}, R_C = \frac{35}{50}$$

▶ Precision = $TP / (TP + FP)$

- Precision A = $40 / (40 + 10 + 5) = 40 / 55 \approx 0.727$
- Precision B = $30 / (5 + 30 + 10) = 30 / 45 \approx 0.667$
- Precision C = $35 / (5 + 10 + 35) = 35 / 50 = 0.700$

▶ Recall = $TP / (TP + FN)$

- Recall A = $40 / (40 + 5 + 5) = 40 / 50 = 0.8$
- Recall B = $30 / (10 + 30 + 10) = 30 / 50 = 0.6$
- Recall C = $35 / (5 + 10 + 35) = 35 / 50 = 0.7$

♦ Step 2: Calculate **Macro Precision and Recall** (average of all classes)

- Macro Precision = $(0.727 + 0.667 + 0.700) / 3 \approx 0.698$
- Macro Recall = $(0.8 + 0.6 + 0.7) / 3 \approx 0.7$

But none of the options match this directly, so let's try **Micro-averaged precision and recall** instead (based on total TP, FP, FN).

[MCQ]

#Q. Assume, you want to cluster 7 observations into 3 clusters using the K-Means clustering algorithm. After the first iteration, the clusters: C1, C2, C3 have the following observations

C1: $\{(1,1), (4,4), (7,7)\} \rightarrow 4,4$

C2: $\{(0,4), (4,0)\} \leftarrow 2,2$

C3: $\{(5,5), (9,9)\} \leftarrow 7,7$

What will be the cluster centroids if you want to proceed for the second iteration?

☒ A

Ct: (4,4), C2: (2,2), C3: (7,7)

☐ B

Ct: (2,2), C2: (0,0), C3: (5,5)

☐ C

Ct: (6,6), C2: (4,4), C3: (9,9)

☐ D

None of these

[MCQ]

#Q. We will use the dataset below to learn a decision tree which predicts if people pass machine learning (Yes or No), based on their previous GPA (High, Medium, or Low) and whether or not studied.

What is the entropy $H(\text{Passed} \mid \text{GPA})$ and $H(\text{Passed} \mid \text{Studied})$?

GPA	Studied	Passed
L ✓	F ✓	F
L	T	T
M	F ✓	F
M	T	T
H	F ✓	T
H	T	T

A $H(\text{Passed} \mid \text{GPA}) = 0.66$ ✓
 $H(\text{Passed} \mid \text{Studied}) = 0.46$

B $H(\text{Passed} \mid \text{GPA}) = 0.46$
 $H(\text{Passed} \mid \text{Studied}) = 0.66$

C $H(\text{Passed} \mid \text{GPA}) = 0.46$
 $H(\text{Passed} \mid \text{Studied}) = 0.92$

D $H(\text{Passed} \mid \text{GPA}) = 0.33$
 $H(\text{Passed} \mid \text{Studied}) = 0.66$

$$\text{GPA} \\ L \rightarrow P_F = 1/2 \quad P_T = 1/2 \quad \mathcal{E} = 1$$

$$M \rightarrow P_F = 1/2 \quad P_T = 1/2 \quad \mathcal{E} = 1$$

$$H \rightarrow P_F = 0 \quad P_T = 1 \quad \mathcal{E} = 0$$

$$H(P/\text{GPA}) \Rightarrow \frac{2 \times 1 + 2 \times 1 + 0 \times 2}{6} = 4/6$$

Studies

$$\begin{array}{l} F \rightarrow P_F = 2/3 \quad P_T = 1/3 \\ T \rightarrow P_F = 0 \quad P_T = 1 \end{array} \quad \mathcal{E}$$

We compute **entropy of Passed** separately for each GPA value, then take the weighted sum:

✓ GPA = L (2 samples: [F, T])

- Passed: 1 T, 1 F → Prob = (0.5, 0.5)
- Entropy = $-[0.5 \log_2(0.5) + 0.5 \log_2(0.5)] = 1$

✓ GPA = M (2 samples: [F, T])

- Passed: 1 T, 1 F → Prob = (0.5, 0.5)
- Entropy = 1

✓ GPA = H (2 samples: [T, T])

- Passed: 2 T → Prob = (1, 0)
- Entropy = $-[1 \times \log_2(1)] = 0$

Now compute weighted average:

$$H(\text{Passed}|\text{GPA}) = \frac{2}{6}(1) + \frac{2}{6}(1) + \frac{2}{6}(0) = \frac{4}{6} = 0.666$$

✓ So, $H(\text{Passed} | \text{GPA}) = 0.66$

Group by **Studied** = F and T:

✅ **Studied = F** → 3 instances: [F, F, T] (GPA L, M, H)

- Passed = F, F, T → 2 F, 1 T → Prob = (1/3, 2/3)

$$Entropy = -[1/3 \log_2(1/3) + 2/3 \log_2(2/3)] \approx 0.918$$

✅ **Studied = T** → 3 instances: [T, T, T]

- Passed = T, T, T → all T → Prob = (1, 0)
- Entropy = 0

Now compute weighted average:

$$H(Passed|Studied) = \frac{3}{6}(0.918) + \frac{3}{6}(0) = 0.459 \approx 0.46$$

[MCQ]

#Q. Consider a binary classification problem with the following dataset:

Sample	Feature 1	Feature 2	Feature 3
A	2	3	0
B	1	5	1
C	4	2	1
D	3	1	0
E	5	4	0
F	2	2	1
G	3	3	1

☒ A 0.514 *n_{sample}*

☐ B 0.486

☐ C 0.353

☐ D 0.65

Calculate the Gini impurity for the entire dataset.

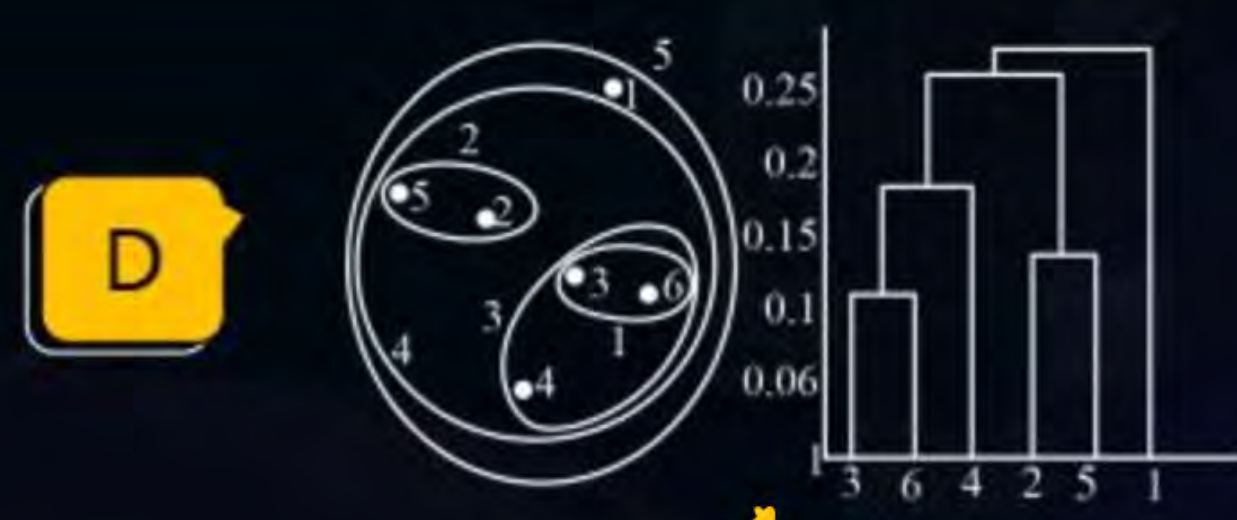
#Q. Given Six points with the following attributes:

Point	x coordinate	y coordinate
p1	0.4005	0.5306
p2	0.2148	0.3854
P3	0.3457	0.3156
p4	0.2652	0.1875
p5	0.0789	0.4139
p6	0.4548	0.3022

Table : X-Y coordinates of six points

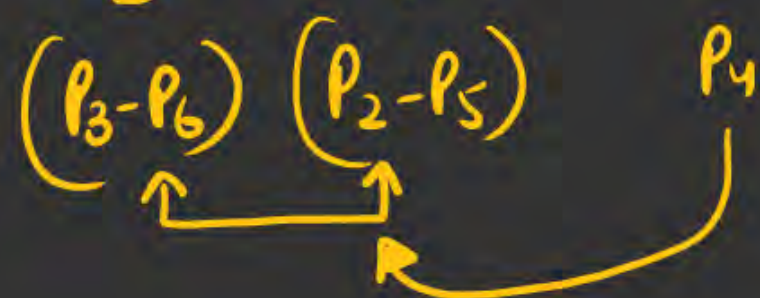
	p1	p2	p3	p4	p5	p6
p1	0.0000	0.2357	0.2218	0.3688	0.3421	0.2347
p2	0.2357	0.0000	0.1483	0.2042	0.1388	0.2540
p3	0.2218	0.1483	0.0000	0.1513	0.2843	0.1100
p4	0.3688	0.2042	0.1513	0.0000	0.2932	0.2216
p5	0.3421	0.1388	0.2843	0.2932	0.0000	0.3921
p6	0.2347	0.2540	0.1100	0.2216	0.3921	0.0000

#Q. Which of the following clustering representations and dendrogram depicts the use of MIN or Single link proximity function in hierarchical clustering?



	p1	p2 ✕	p3 ✕	p4 ✕	p5 ✕	p6
p1	0.0000	0.2357	0.2218	0.3688	0.3421	0.2347
p2	0.2357	0.0000	0.1483	0.2042	0.1388 ✓	0.2540
p3	0.2218	0.1483	0.0000	✓0.1513	0.2843	✓0.1100
p4	0.3688	0.2042	0.1513	0.0000	0.2932	0.2216 ✓
p5	0.3421	0.1388	0.2843	0.2932	0.0000	0.3921
p6	0.2347	0.2540	✓0.1100	0.2216	0.3921	0.0000

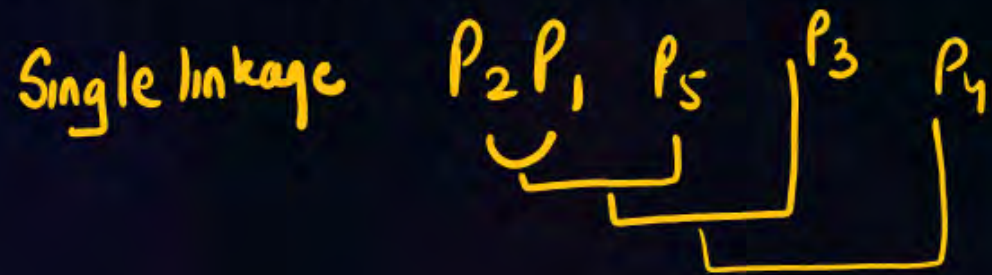
Single linkage

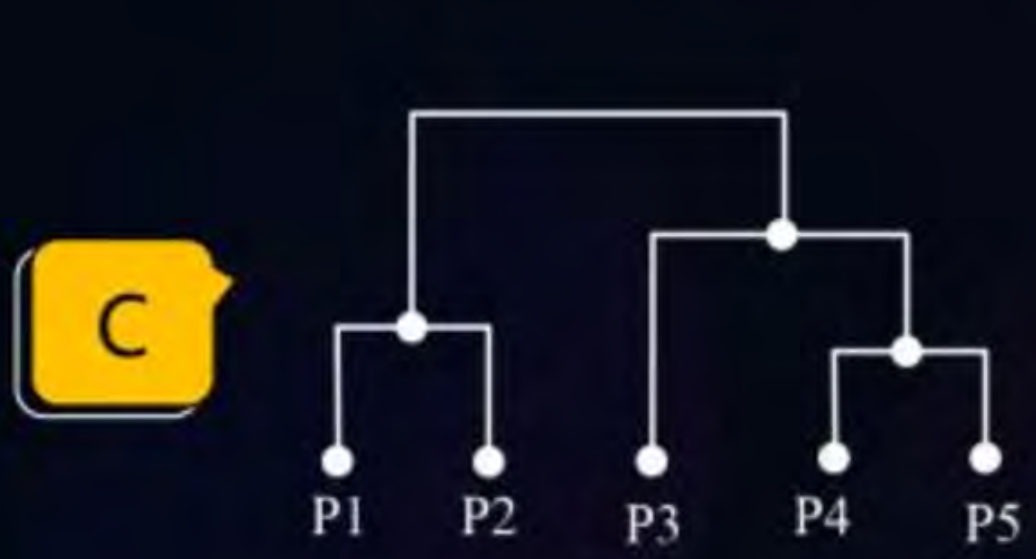
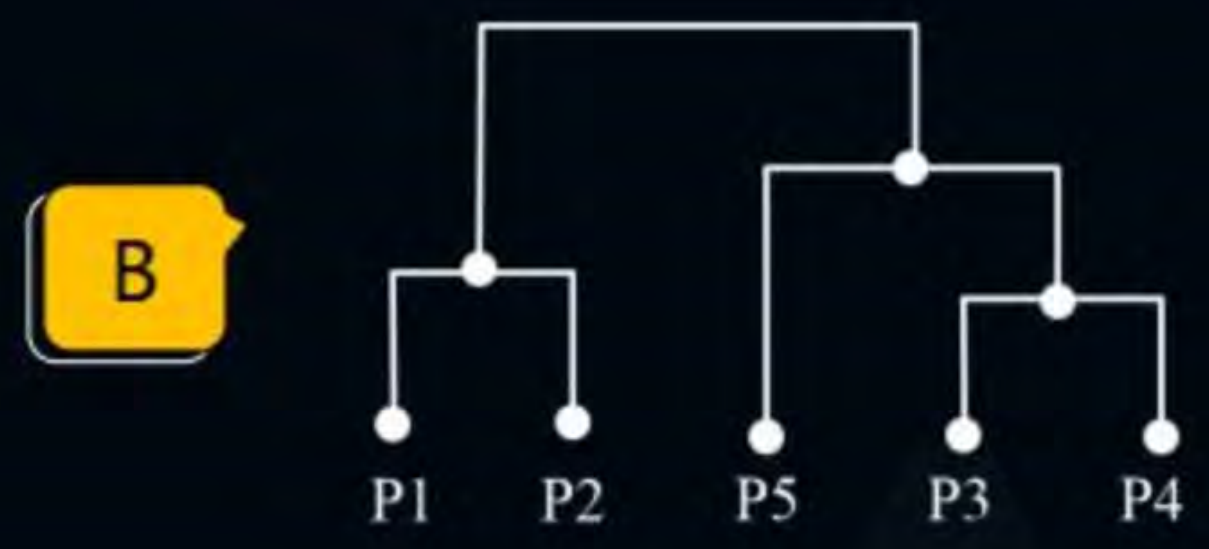
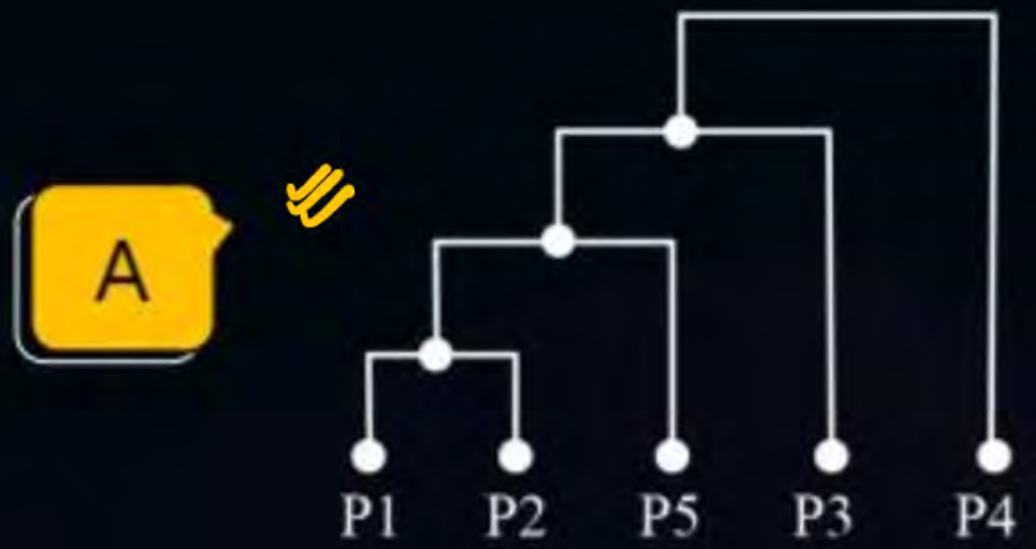


#Q. Consider the distance matrix in Table1 to perform the complete link hierarchical clustering.

	P1	P2	P3	P4	P5
P1	0.00	0.10	0.41	0.55	0.35
P2	0.10	0.00	0.64	0.47	0.98
P3	0.41	0.64	0.00	0.44	0.85
P4	0.55	0.47	0.44	0.00	0.76
P5	0.35	0.98	0.85	0.76	0.00

Which ONE of the following is the correct representation of the clusters produced using dendrogram?





#Q. Given the following 8 data points in a 2-dimensional space:

Point	Coordinates (x, y)
A1	(2, 10)
A2	(2, 5)
A3	(8, 4)
A4	(5, 8)
A5	(7, 5)
A6	(6, 4)
A7	(1, 2)
A8	(4, 9)

Assume that initial centroid:

Centroid 1 (C1): A1 = (2, 10)

Centroid 2 (C2): A4 = (5, 8)

Centroid 3 (C3): A7 = (1, 2)

Cluster these points into 3 clusters using the K-means algorithm.

Choose the correct option.

Point	Coordinates (x, y)
A1	(2, 10) ✓ C ₁
A2	(2, 5) C ₃
A3	(8, 4) C ₂
A4	(5, 8) C ₂
A5	(7, 5) C ₂
A6	(6, 4) C ₂
A7	(1, 2) C ₃
A8	(4, 9) C ₂

C₁ 2, 10

0

5

$$\sqrt{6^2 + 6^2}$$

$$\sqrt{5^2 + 5^2}$$

$$\sqrt{4^2 + 6^2}$$

$$\sqrt{2^2 + 1^2}$$

C₂ 5, 8

$$\sqrt{9 + 9}$$

$$\sqrt{3^2 + 4^2}$$

0

$$\sqrt{2^2 + 3^2}$$

$$\sqrt{1^2 + 4^2}$$

$$\sqrt{1^2 + 1^2}$$

C₃ 1, 2

$$\sqrt{1 + 9}$$

$$\sqrt{7^2 + 2^2}$$

$$\sqrt{6^2 + 3^2}$$

$$\sqrt{5^2 + 2^2}$$

$$\sqrt{3^2 + 7^2}$$

A

Cluster 1: {A1, A2}

Cluster 2: {A3, A4, A5, A6, A8}

Cluster 3: {A7}

B

Cluster 1: {A1}

Cluster 2: {A3, A4, A5, A6, A8}

Cluster 3: {A2, A7}

C

Cluster 1: {A1, A2}

Cluster 2: {A3, A4, A5, A8}

Cluster 3: {A6, A7}

D

Cluster 1: {A1}

Cluster 2: {A4, A5, A6, A8}

Cluster 3: {A2, A3, A7}

[MCQ]

#Q. We will use the dataset below to learn a decision tree which predicts if people pass machine learning (Yes or No), based on their previous GPA (High, Medium, or Low) and whether or not they studied

GPA	Studied	Passed
L	F	F
L	T	T
M	F	F
M	T	T
H	F	T
H	T	T

What is the entropy $H(\text{Passed})$?

A 0.46

C 0.92 ✓

B 0.63

D 0.23

✓ Step 1: Count occurrences

Total = 6 samples

- Passed = T (True): 4
- Passed = F (False): 2

So the probabilities are:

- $P(T) = \frac{4}{6} = 0.667$
- $P(F) = \frac{2}{6} = 0.333$

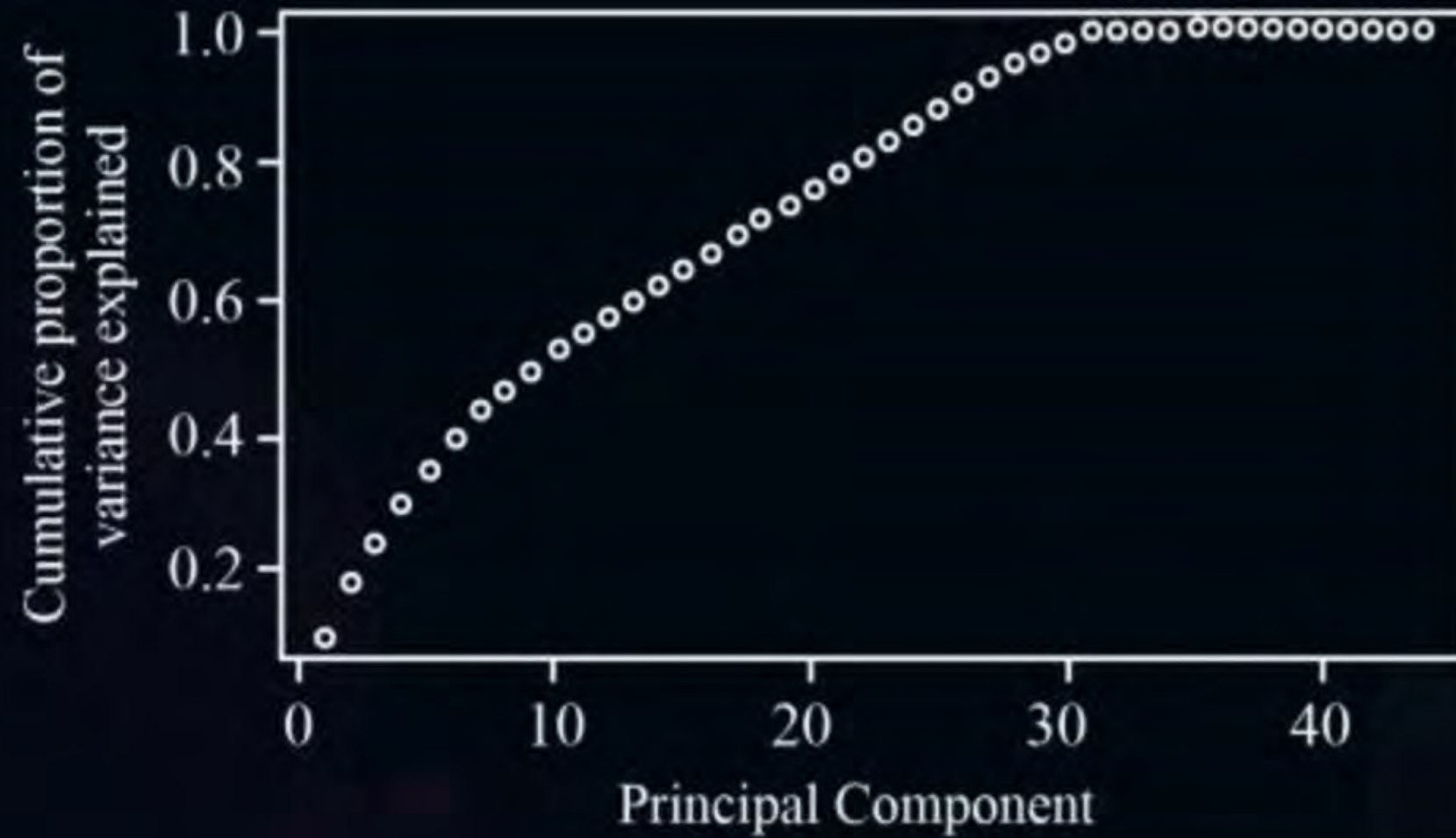
✓ Final Answer: Option C

$$H(\text{Passed}) = 0.92$$

✓ Step 2: Use Entropy Formula

$$\begin{aligned}
 H(\text{Passed}) &= -\sum p(x) \cdot \log_2 p(x) \\
 &= -(0.667 \cdot \log_2(0.667) + 0.333 \cdot \log_2(0.333))
 \end{aligned}$$

#Q. What are the optimal number of principal components in the below figure?



A 10

B 20

C 30

D 40

[MCQ]

#Q. Euclidean distance based k-means clustering algorithm was run on a dataset of 100 points with $k = 3$. If the points $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ are both part of cluster 3, then which ONE of the following points is necessarily also part of cluster 3?

A $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$

B $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$

C $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$

D $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

#Q. Problem Given the dataset {a, b, c, d, e) and the following distance matrix, construct a dendrogram by complete linkage hierarchical clustering using the agglomerative method.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

	a	b	ce	d
a	0	9✓	11✓	6
b	9	0	10	5
ce	11	10	0	9
d	6	5✓	9✓	0

	ad	b	ce
ad	0	9	11
b	9	0	10
ce	11	10	0

$((ce) (ad) b) ✓$

↑ ↑ ↑

[MCQ]

- #Q. In solving a n-armed bandit problem, we decide to use the-greedy algorithm. We have the following two choices for the ϵ parameter, $\epsilon = 0.1$ and $\epsilon = 0.01$. If we run the algorithm for a very long time, and want to maximize the cumulative reward, which value of ϵ should we use?
- Suppose we want to find an optimal arm in the fastest time. In this situation, which value of ϵ should we use?
- A** for maximum reward use $\epsilon = 0.1$ for minimizing the time required to find an optimal arm, use $\epsilon = 0.01$
 - B** for maximum reward use $\epsilon = 0.01$ for minimizing the time required to find an optimal arm, use $\epsilon = 0.1$
 - C** for both cases, use $\epsilon = 0.1$
 - D** for both cases, use $\epsilon = 0.01$

Skip.

[MCQ]

#Q. Consider the following one dimensional data set: 12, 22, 2, 3, 33, 27, 5, 16, 6, 31, 20, 37, 8, 18. Given $k = 3$ and initial cluster centers to be 5, 6 and 31, what are the final cluster centers obtained on applying the k-means algorithm?

→ 1st it ⇒ $\left(\frac{10}{3}, 12, \underline{\underline{28.33}}\right)$

A 5, 18, 30

C 6, 19, 32

B 5, 18, 32

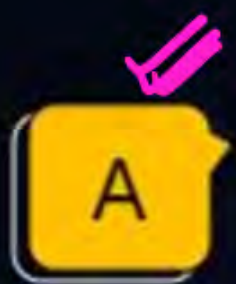
D 4.8, 17.6, 32

31
33
37

	5	6	31
2	3	4	
3	2	3	
5	0	1	
6	1	0	
8	3	2	
12	7	6	
16	11	10	
18	13	12	13
20	15	14	11
22			
27			

#Q. For the similarity matrix given below, show the hierarchy of clusters created by the single link clustering algorithm.

	P1	P2	P3	P4	P5	P6
P1	1.0000	0.7895	0.1579	0.0100	0.5292	0.3542
P2	0.7895	1.0000	0.3684	0.2105	0.7023	0.5480
P3	0.1579	0.3684	1.0000	0.8421	0.5292	0.6870
P4	0.0100	0.2105	0.8421	1.0000	0.3840	0.5573
P5	0.5292	0.7023	0.5292	0.3840	1.0000	0.8105
P6	0.3542	0.5480	0.6870	0.5573	0.8105	1.0000



[MCQ]

#Q. In the figure below, if you draw a horizontal line on y-axis for $y=2$. What will be the number of clusters formed?



A 1

B 2 ✓

C 3

D 4

[MCQ]

#Q. Assume, you want to cluster 7 observations into 3 clusters using K-Means clustering algorithm. After first iteration the clusters: C1, C2, C3 has the following observations:

C1: $\{(1,1), (4,4), (7,7)\}$

C2: $\{(0,4), (4,0)\}$

C3: $\{(5,5), (9,9)\}$

done

What will be the cluster centroids if you want to proceed for second iteration?

A C1: (4,4), C2: (2,2), C3: (7,7)

B C1: (2,2), C2: (0,0), C3: (5,5)

C C1: (6,6), C2: (4,4), C3: (9,9)

D None of these

#Q. If two variables V_1 and V_2 are used for clustering. Which of the following are true for K-means clustering with $k=3$?

- ✓ 1. If V_1 and V_2 has a correlation of 1, the cluster centroids will be in a straight line
2. If V_1 and V_2 has a correlation of 0, the cluster centroids will be in straight line

Choose the correct answer?



A 1 Only ✓

B 2 Only

C Both 1 and 2

D None of the above

[MCQ]

#Q. Which of the following clustering algorithms is the most sensitive to outliers?

☒ A K-means clustering algorithm

☐ B K-medians clustering algorithm

☐ C K-modes clustering algorithm

☐ D None of the above

Centroid $\Rightarrow \left(\frac{\sum x}{N} \right)$

not affected by outliers

$(4,4)$

$(2,4) (4,3) (4,4) (6,8) (5,6) (3,2)$

2 3 4 4 5 6
2 3 4 4 6 8
4

1. K-medians Clustering

Goal:

To minimize the **sum of Manhattan distances** (L1 norm) between the data points and the cluster center.

Centroid in K-medians:

- The **centroid** is the **median** of all points in a cluster.
- It is computed **independently for each feature**.

Example:

Let's say we have the following points in a cluster:

makefile

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Points: (2, 5), (4, 8), (6, 9)

To find the centroid:

- For **X-coordinates**: median of (2, 4, 6) $\rightarrow$ 4
- For **Y-coordinates**: median of (5, 8, 9) $\rightarrow$ 8
- Centroid = (4, 8)

2. K-modes Clustering

Goal:

Used for **categorical data**; aims to minimize the **number of mismatches** (not distance) in categories.

Centroid in K-modes:

- The **centroid** is the **mode** (most frequent value) of each categorical attribute within the cluster.

Example:

Let's say we have three categorical data points:

CS5

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```
["Red", "S", "Cotton"]  
["Blue", "S", "Cotton"]  
["Red", "M", "Cotton"]
```

} (Red, S, Cotton) ✓✓✓

We calculate the **mode** for each column:

- Color: "Red" (appears twice)
- Size: "S" and "M" both once, but if tie → pick randomly or first encountered
- Material: "Cotton" (all same)
- ♦ Centroid = ["Red", "S", "Cotton"]

[MCQ]

#Q. You want to cluster 7 points into 3 clusters using the k-Means Clustering algorithm. Suppose after the first iteration, clusters C_1 , C_2 and C_3 contain the following two-dimensional points:

C_1 contains the 2 points: $\{(0,6), (6,0)\}$ ✓

C_2 contains the 3 points: $\{(2,2), (4,4), (6,6)\}$

C_3 contains the 2 points: $\{(5,5), (7,7)\}$

What are the cluster centers computed for these 3 clusters?

A

$C_1: (3,3)$, $C_2: (4,4)$, $C_3: (6,6)$ ✓

B

$C_1: (3,3)$, $C_2: (6,6)$, $C_3: (12,12)$

C

$C_1: (6,6)$, $C_2: (12,12)$, $C_3: (12,12)$

D

$C_1: (0,0)$, $C_2: (48,48)$, $C_3: (35,35)$

[NAT]

#Q. Given the dataset:

$(1,2), (2,3), (3,1), (8,8), (9,9), (10,10), (6,7)$

Assuming the initial centroids for $K = 3$ means clustering to be: $C_1 = (2, 3)$ $C_2 = (3, 1)$ $C_3 = (9, 9)$

✓ One iteration of the Expectation-Maximization Algorithm for K-means clustering will update C_1 to:

$$\Rightarrow C_1 \Rightarrow \left(\frac{3}{2}, \frac{5}{2} \right)$$

$$\begin{array}{l} \leftarrow \begin{cases} C_1 \leftarrow 1,2 \\ C_1 \leftarrow 2,3 \\ C_2 \leftarrow 3,1 \end{cases} \end{array}$$

	$\frac{2,3}{\sqrt{1^2+1^2}}$	$\frac{3,1}{\sqrt{2^2+1^2}}$	$9,9$
C_3			
$\left\{ \begin{array}{l} 8,8 \\ 9,9 \\ 10,10 \\ 6,7 \end{array} \right\}$	$\sqrt{4^2+4^2}$	$\sqrt{3^2+6^2}$	$\sqrt{3^2+2^2}$

[NAT]

- #Q. Consider the pairwise distances between 5 points given as matrix. Since the matrix is symmetric, we have given only the lower triangle. Find the final Clustering using single linkage.

	1	2	3	4	5
1	0				
2	2	0			
3	6	3	0		
4	10	<u>9</u>	7	0	
5	9	8	5	4	0

$$\left(\left((1,2) \right) 3 \right) (4,5)$$

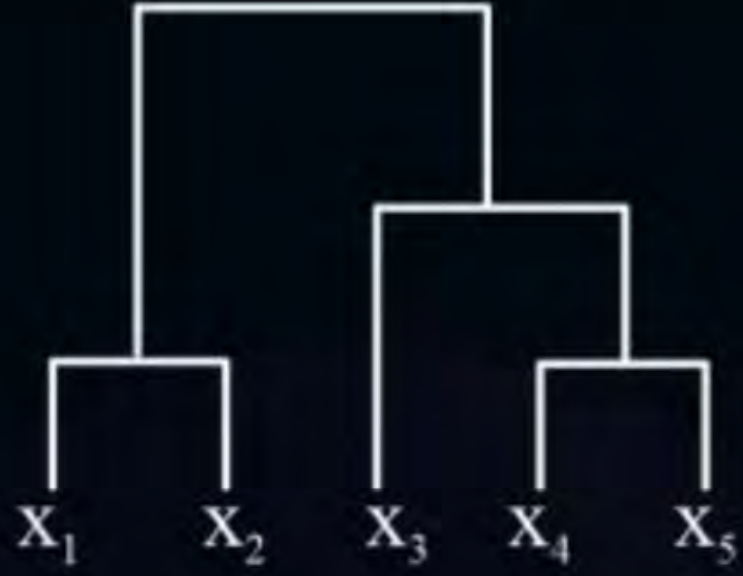
[MCQ]

#Q. Consider the table below, where the $(i, j)^{\text{th}}$ element of the table is the distance between points x_i and x_j . Single linkage clustering is performed on data points, x_1, x_2, x_3, x_4, x_5 .

	x_1	x_2	x_3	x_4	x_5
x_1	0				
x_2	1	0			
x_3	4	3	0	2	5
x_4	3	5	2	0	
x_5	6	3	5	1	0

Which ONE of the following is the correct representation of the clusters produced?

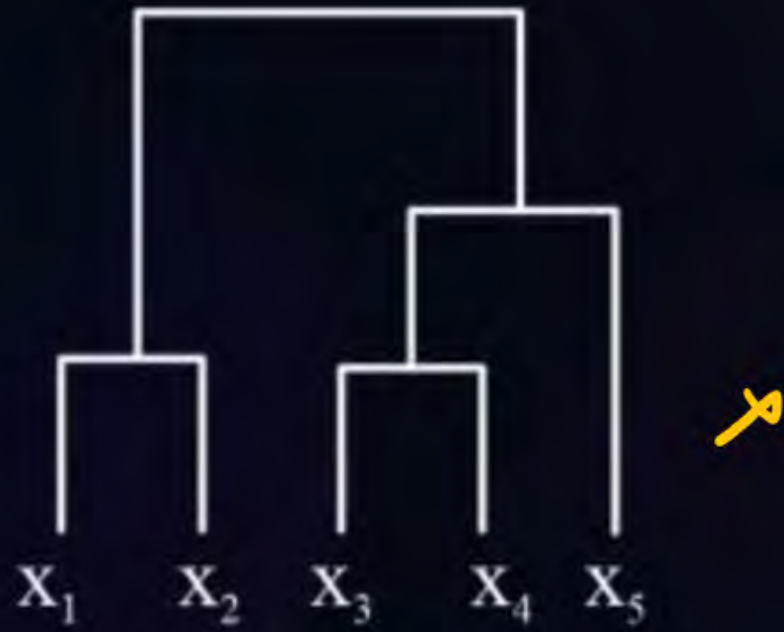
A



B



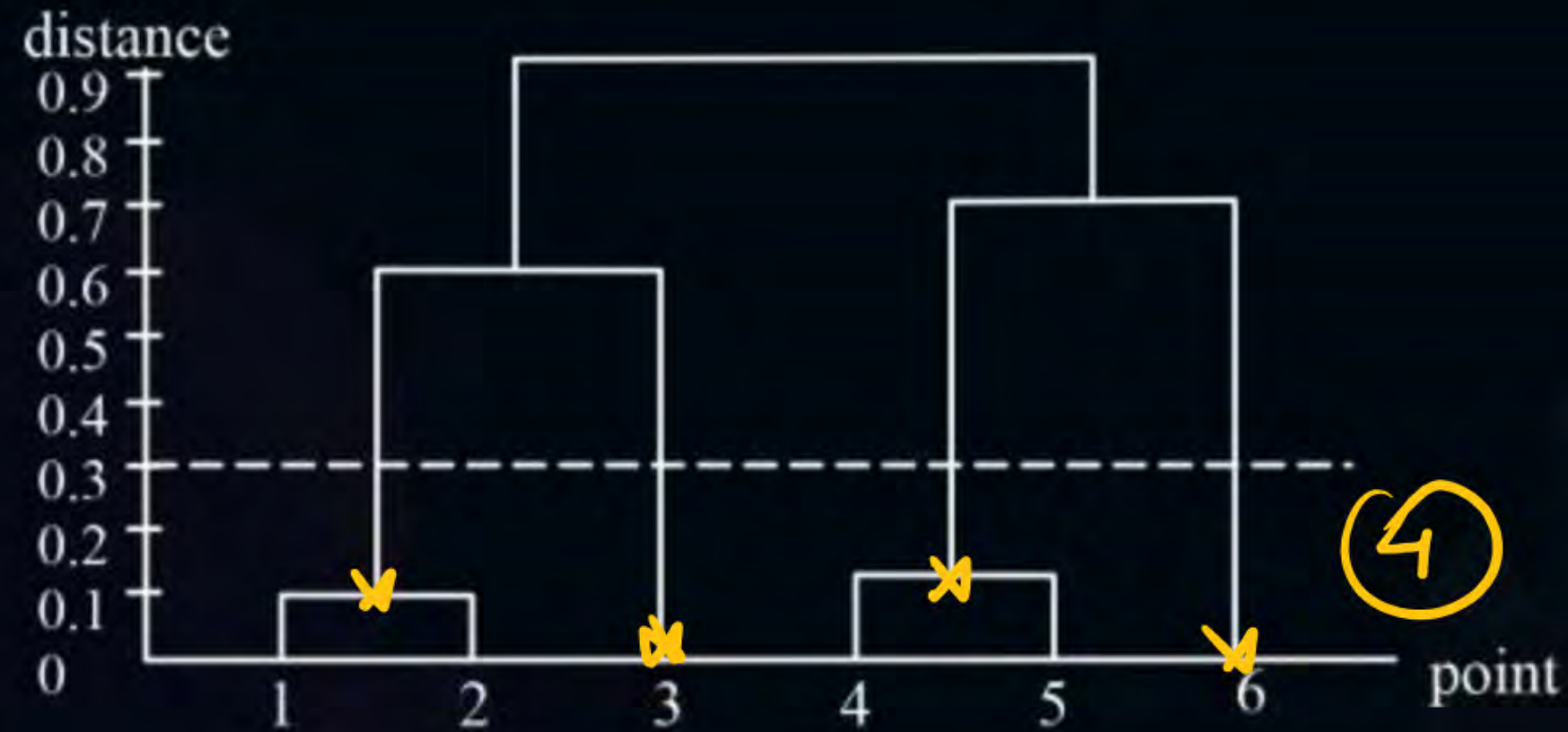
C



D



#Q. Consider the following dendrogram. How many clusters would be reported if were to slice at the dotted line?



[MCQ]

#Q. Consider n feature vectors $x \in \mathbb{R}^d$ that need to be projected into a lower-dimensional space $\mathbb{R}^k$ using k principal component basis vectors. The projection matrix W of size $d \times k$ consists of these orthonormal principal component vectors.

Which of the following expressions correctly represents the reconstruction error?

☒ A $\|x - WW^T x\|_2^2$

$$\|x - (\omega^T x) \omega\|^2$$

☐ B $\|x - W^T x\|_2^2$

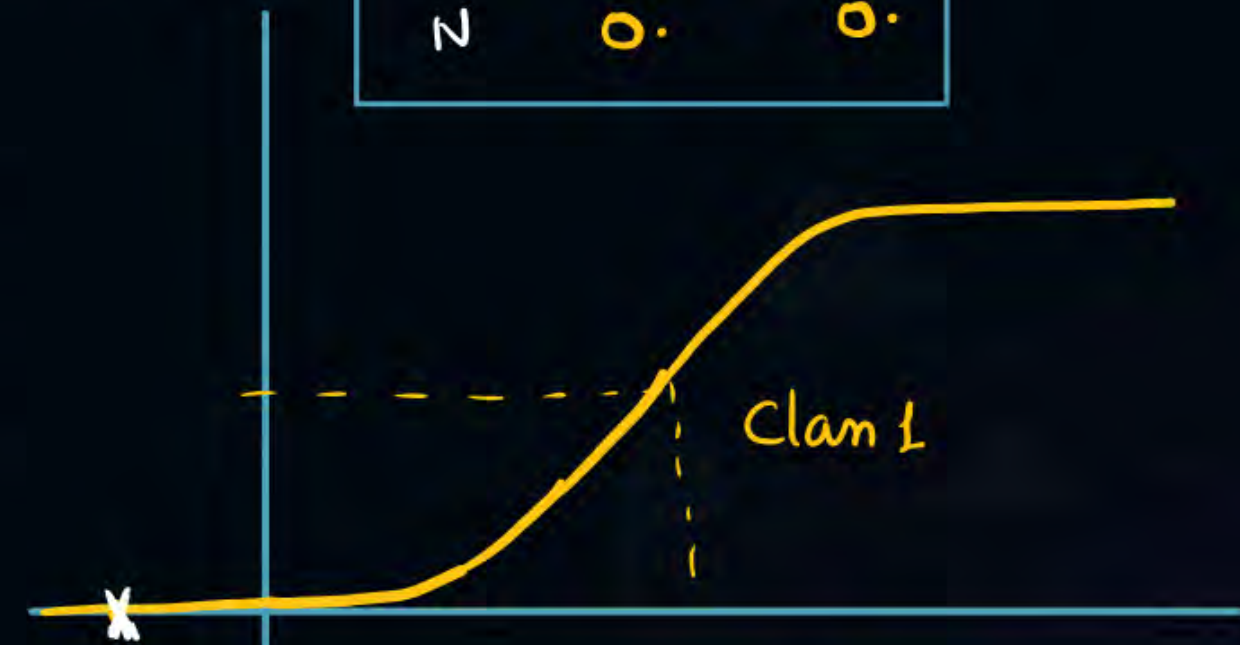
☐ C $\|x - W^T W x\|_2^2$

☐ D $\|x - W x\|_2^2$

#Q. In the context of binary classification, if we continuously decrease the classifier threshold, how do the True Positive Rate (TPR) and False Positive Rate (FPR) behave?

* (all data predict = Positive)

Pred	Actual	
	P	N
P	50	50
N	0	0



all clan 1 $\Rightarrow$ $TPR = \frac{TP}{\text{actual } P} = \frac{50}{50} = 1$, $FPR = \frac{FP}{\text{Actual } N} = \frac{50}{50} = 1$

- A TPR increases while FPR decreases
- B TPR decreases while FPR increases
- ☒ C Both TPR and FPR increase
- D Both TPR and FPR decrease

[Common Data]

Let's do principal components analysis (PCA)! Consider this sample of six points $x_i \in \mathbb{R}^2$.

$$\left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\} \xrightarrow{\text{label}} \begin{matrix} 1 \\ 1 \end{matrix}$$

#Q. Compute the mean of the sample points and write the centered design matrix X .

$$\left(\begin{matrix} -1 & -1 & 0 & 0 & 1 & 1 \\ -1 & 0 & -1 & 1 & 0 & 1 \end{matrix} \right)$$

$$X = \begin{bmatrix} -1 & -1 & 0 & 0 & 1 & 1 \\ -1 & 0 & -1 & 1 & 0 & 1 \end{bmatrix}_{2 \times 6} \quad X^T = \begin{bmatrix} -1 & -1 \\ -1 & 0 \\ 0 & -1 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$C = \frac{1}{6} \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$$

Let's do principal components analysis (PCA)! Consider this sample of six points $x_i \in \mathbb{R}^2$.

$$\left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\}.$$

#Q. Find all the principal components of this sample.
Write them as unit vectors.

$$C = \frac{1}{6} \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \quad \begin{pmatrix} 4-\lambda & 2 \\ 2 & 4-\lambda \end{pmatrix} \Rightarrow \begin{aligned} (4-\lambda)^2 - 4 &= 0 \\ 16 + \lambda^2 - 8\lambda - 4 &= 0 \\ \lambda^2 - 8\lambda + 12 &= 0 \\ \lambda &= 6, 2 \end{aligned}$$

$$\underline{\underline{\lambda = 6}}$$

$$\begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$-2x_1 + 2x_2 = 0$$

$$(x_1 = x_2)$$

$$\Downarrow \Rightarrow \begin{bmatrix} x_{f2} \\ x_{f2} \end{bmatrix} \checkmark$$

Capture $6/8 \times 100$

$$\lambda = 2$$

$$\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$2x_1 + 2x_2 = 0$$

$$x_1 = -x_2$$

$$\begin{bmatrix} x_{f2} \\ -x_{f2} \end{bmatrix}$$

$$2/8 \times 100.$$

Let's do principal components analysis (PCA)! Consider this sample of six points $x_i \in \mathbb{R}^2$.

$$\left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\}.$$

- #Q. • Which of those two principal component would be preferred if you use only one?
- What information does the PCA algorithm use to decide that one principal component is better than another? $\lambda=6, 2$
 - From an optimization point of view, why do we prefer that one?

Let's do principal components analysis (PCA)! Consider this sample of six points $x_i \in \mathbb{R}^2$.

$$\left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\}.$$

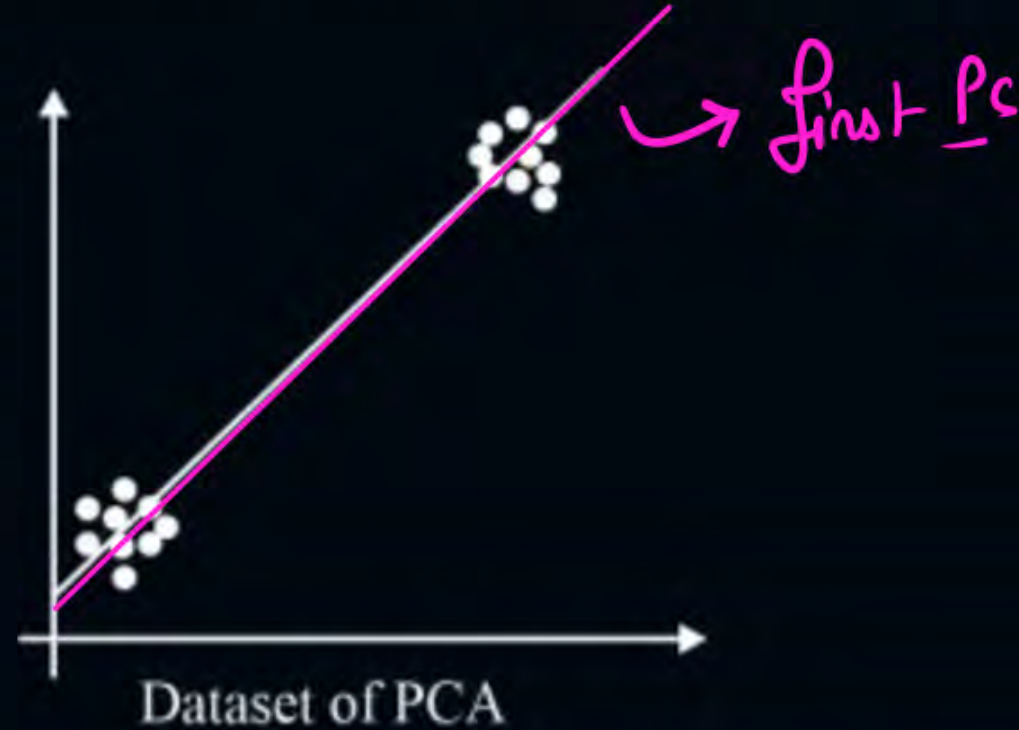
#Q. Compute the vector projection of each of the original sample point (not the centered sample points) onto your preferred principal component. By “vector projection” we mean that the projected points are still in $\mathbb{R}^2$. (Don't just give us the principal coordinate give us the projected point).

$$w = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, \quad \textcircled{1} (w^T x \cdot w) \Rightarrow \textcircled{0}, \textcircled{2} \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$$

$$\textcircled{3} \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}, \quad \textcircled{4}, \textcircled{5} \frac{3}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 3/2 \\ 3/2 \end{bmatrix}, \quad \textcircled{6} \begin{pmatrix} 4/2 \\ 4/2 \end{pmatrix}$$

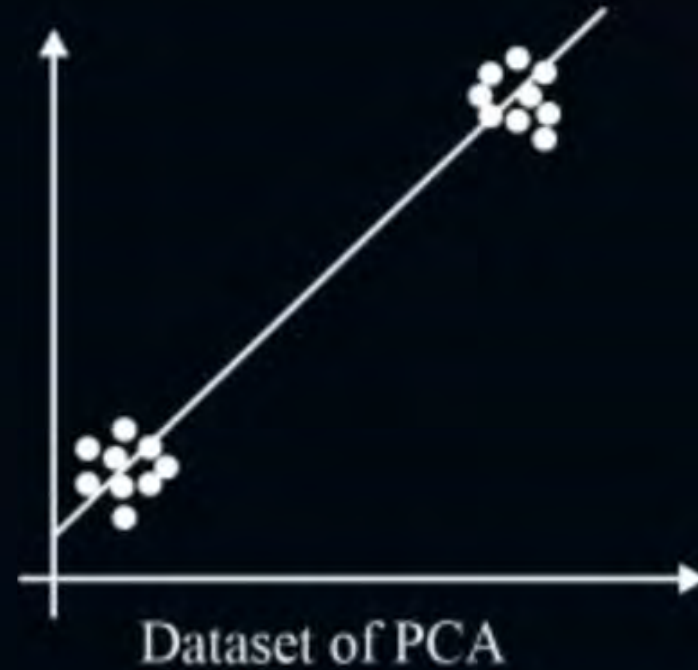
[Common Data]

Suppose you were to use PCA of the following dataset.



#Q. On the figure above, draw the line that is the direction on the first principal component of the data.

Suppose you were to use PCA of the following dataset.

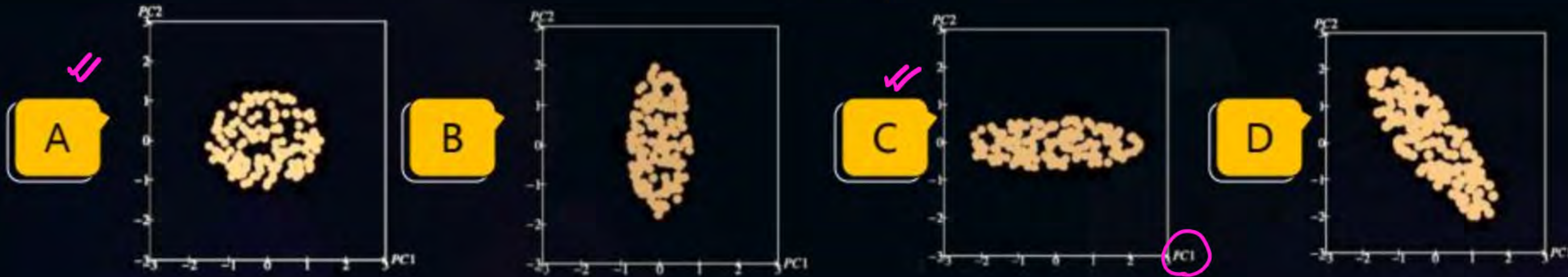


#Q. Justify your answer to part (a) in one sentence.

[MCQ]

#Q. In each plot below, the data is projected onto two (unit-length) principal component vectors. We say that a plot is "valid" if the x-coordinate (written as "PC1") would be the first chosen principal coordinate and the y-coordinate (written as "PC2") would be the second chosen principal coordinate. Which plots are valid?

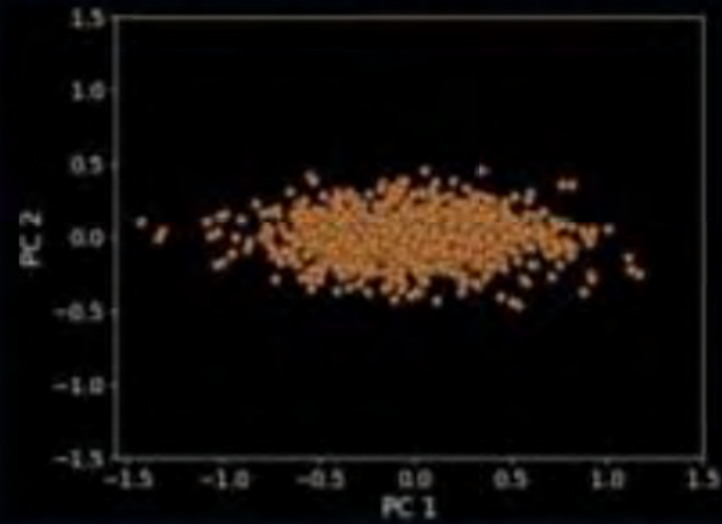
PC1 capture more variance than PC2



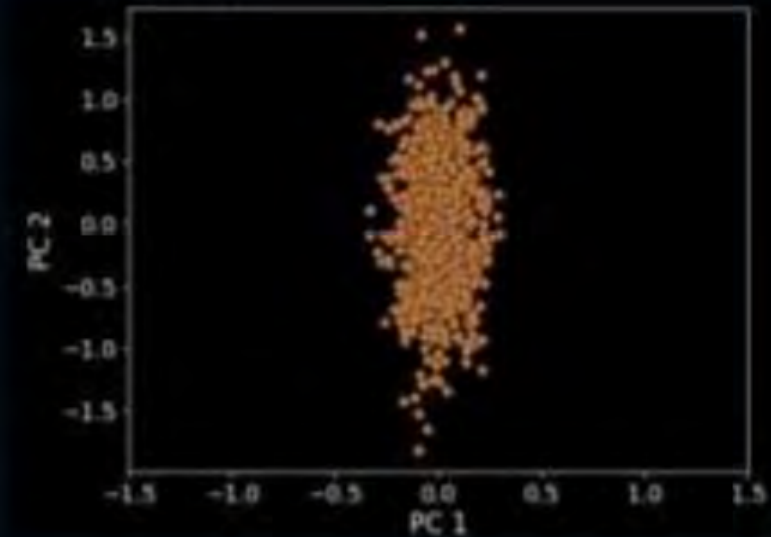
[MCQ]

#Q. Suppose we project a new matrix A onto the directions of its first two principal components. Which of the following plots could possibly display the projected data with the first PC plotted along the horizontal axis and the second PC plotted along the vertical axis? Select all that apply.

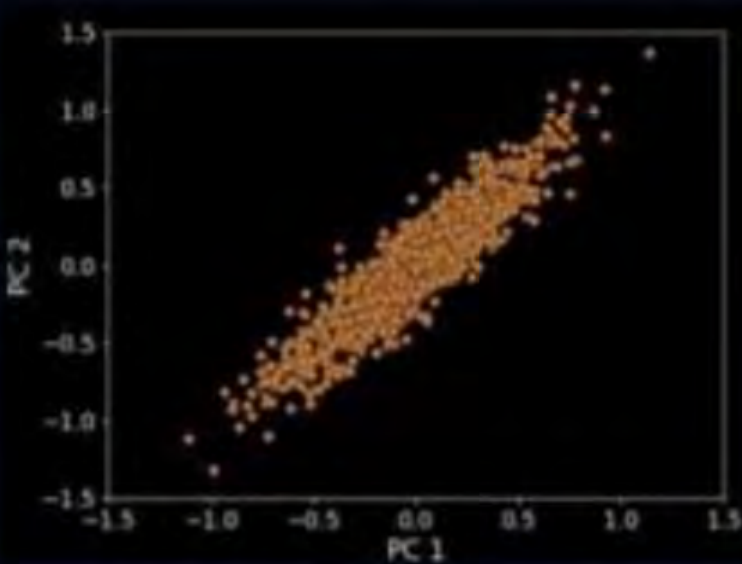
A



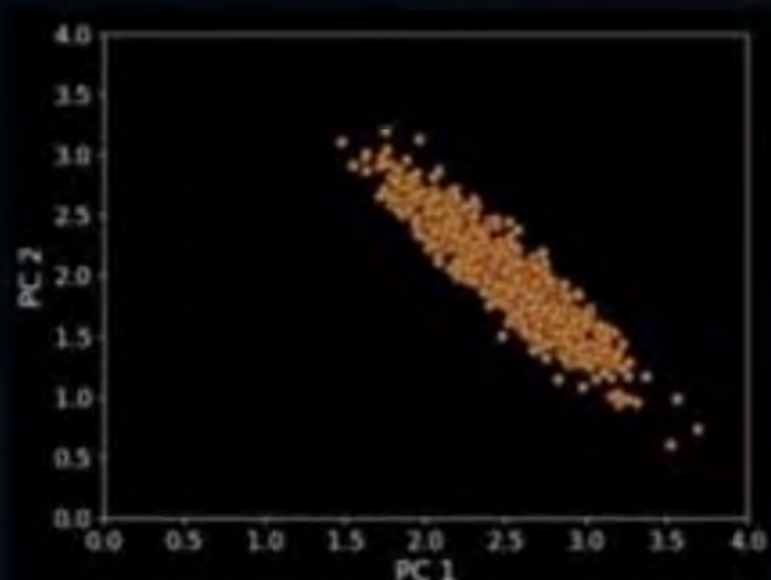
B



C



D



[MCQ]

#Q. Suppose we are given three datasets A,B and $C \in \mathbb{R}^{100 \times 2}$ i.e. each dataset consists of 100 data points in two dimensions. We visualize the datasets using scatterplots, labelled A Plot B, and Plote C, respectively:



If we applied PCA to each of the above dataset and used the first two principal component, which dataset(s) would have to lowest reconstruction error?

A

Dataset A ✓

B

Dataset B ✓

C

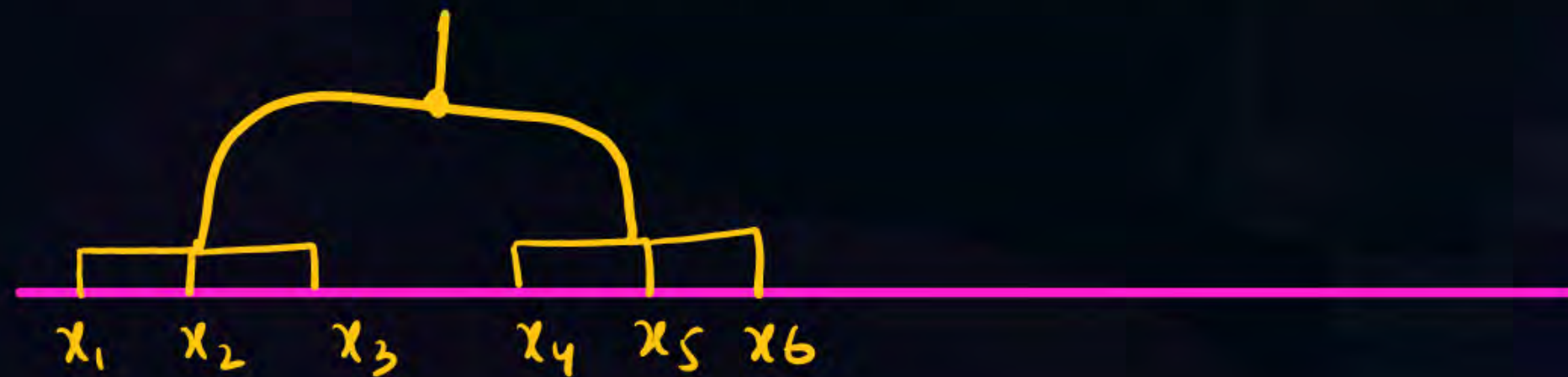
Dataset C ✓

D

Cannot be determined

#Q. Perform agglomerative Hierarchical Clustering using single linkage as the cluster distance measure. Draw the associated tree.

	$\{x_1\}$	$\{x_2\}$	$\{x_3\}$	$\{x_4\}$	$\{x_5\}$	$\{x_6\}$
$\{x_1\}$	0	1	2	1.5	1.5	1.5
$\{x_2\}$	1	0	1	2.5	1.5	2.5
$\{x_3\}$	2	1	0	3.5	2.5	3.5
$\{x_4\}$	1.5	2.5	3.5	0	1	1
$\{x_5\}$	1.5	1.5	2.5	1	0	1
$\{x_6\}$	1.5	2.5	3.5	1	1	0



[NAT]

#Q. (Hierarchical Clustering) Consider six points in 2D, some of them overlap. We start with three clusters (rather than from scratch):

Cluster 1 = (0,0), (1,1), (2,2)

Cluster 2 = (0,1), (1,0)

Cluster 3 = (0,0)

With Euclidean distance and complete linkage, which clusters are merged in the next step of hierarchical clustering?

$C_1 \cap C_2$
 $(0,0)$ 1,1
 $(1,1)$ 1,1
 $(2,2)$ $\sqrt{5}, \sqrt{5}$ ✓

$$C_1 C_3 = \sqrt{4^2 + 4^2} \quad \checkmark$$

$$C_2 C_3 = 1$$

$C_2 C_3$ ✓

[MCQ]

#Q. Perform hierarchical clustering with single linkage in one-dimensional space on the following points: -5, -1, 0, 3, 9, 11. Draw the clustering tree.

Sol

Iteration 1: group $A = (-1, 0)$: distance 1

Iteration 2: group $B = (9, 11)$: distance 2

Iteration 3: group $C = (A, 3)$: distance 3 between 3 and the element 0 in A

Iteration 4: group $D = (-5, C)$: distance 4 between -5 and the element -1 in C

Iteration 5: group $E = (D, B)$: distance 6 between elements 3 and 9.

No tree graph, but correct steps: 7 points

Almost correct graph and solution, minor errors: 7 points

Sol $\Rightarrow$



[NAT]

#Q. There are five points in one-dimensional space: $a = 1.5$, $b = 2$, $c = 6$, $d = 7$, $e = 9$. Perform Hierarchical Agglomerative Clustering with Single Linkage, with the following extra constraint: at no time should c and d be in the same cluster. Complete the resulting clustering tree diagram (i.e., the dendrogram).

	a	b	c	d	e
a	1.5	2	6	7	9
b	1.5	2	4	5	7
c	4.5	4	6	1	3
d	5.5	5	1	7	2
e	7.5	7	3	2	9

$(c(a,b))(d,e)$ 2 clusters



THANK - YOU